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Course Code : EE 4121

VLSI

Static & Dynamic Flip-Flops

Static Flip Flops

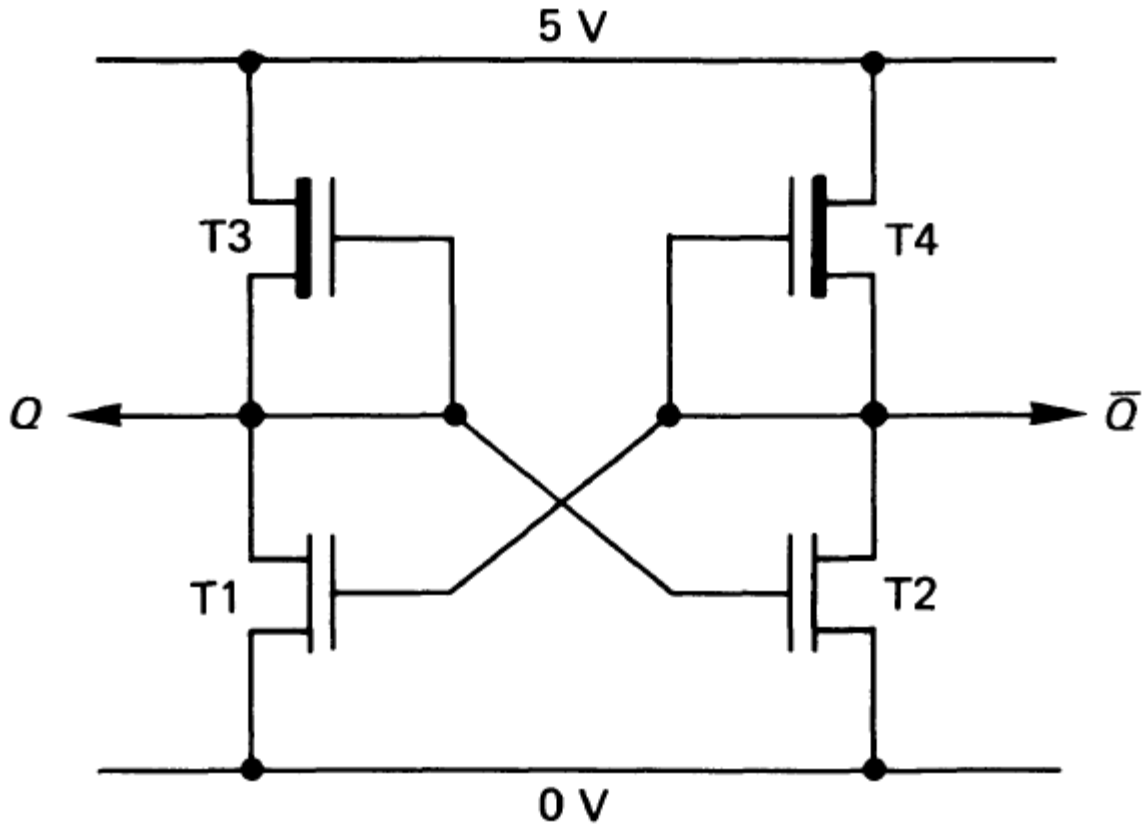
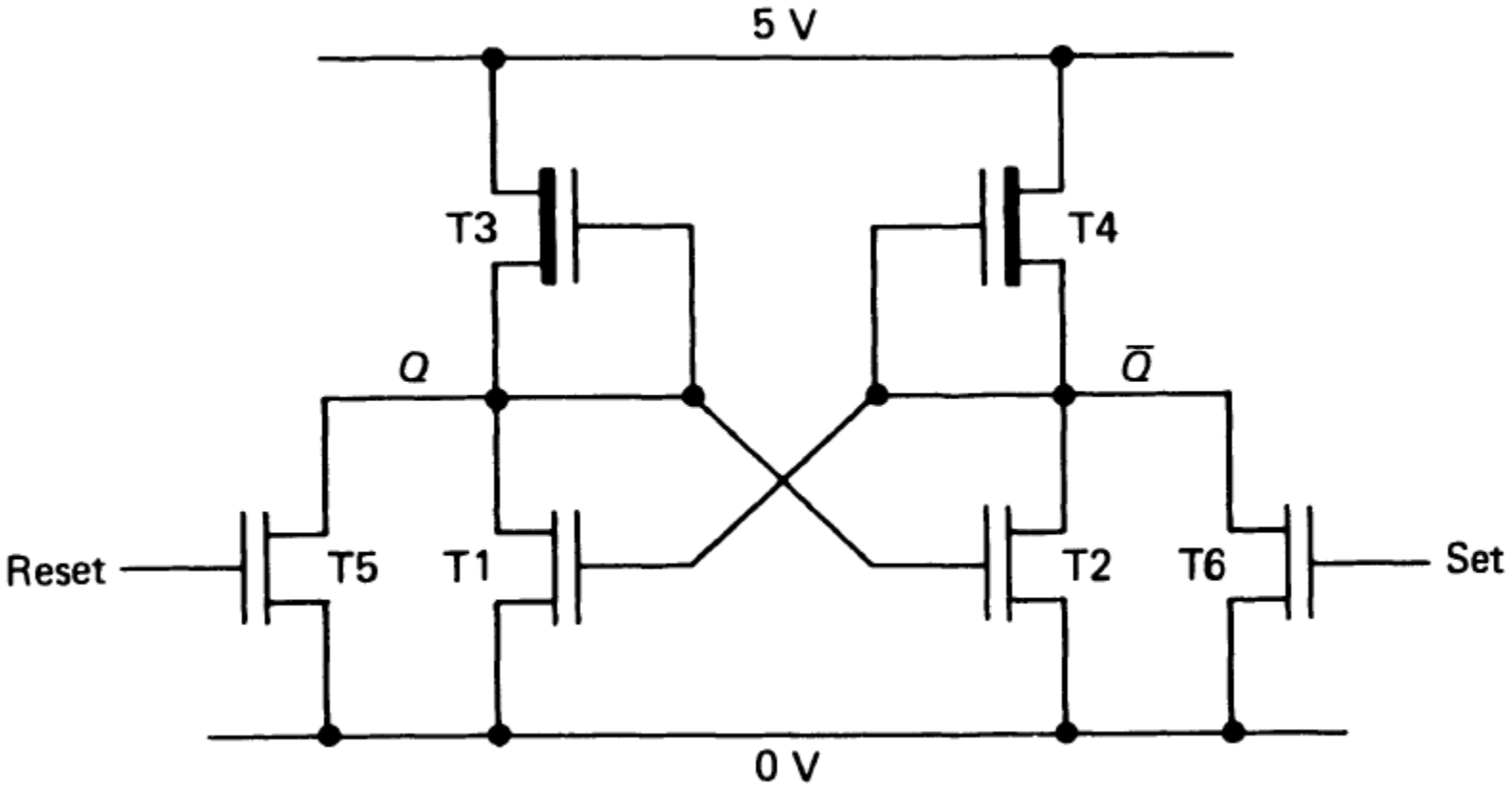


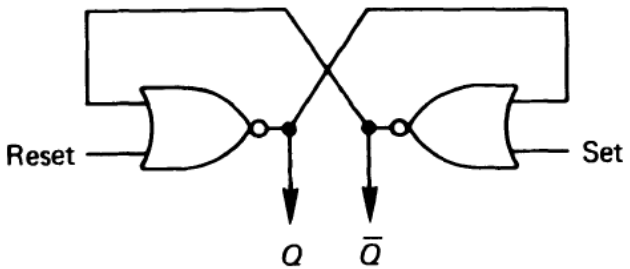
Figure 4.9 Basis of static (NMOS) flip flop

- T1 is high at 5 V => T1 is on => *Q* is low
- *Q* is low => T2 is off => $\overline{Q}$ is high

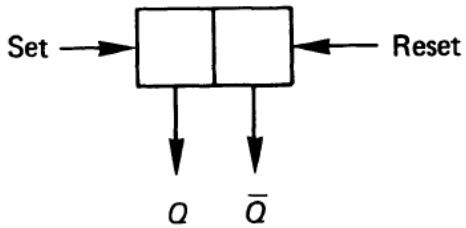
Similar for high value at the gate of T2.

Set-Reset flip flop





(b)

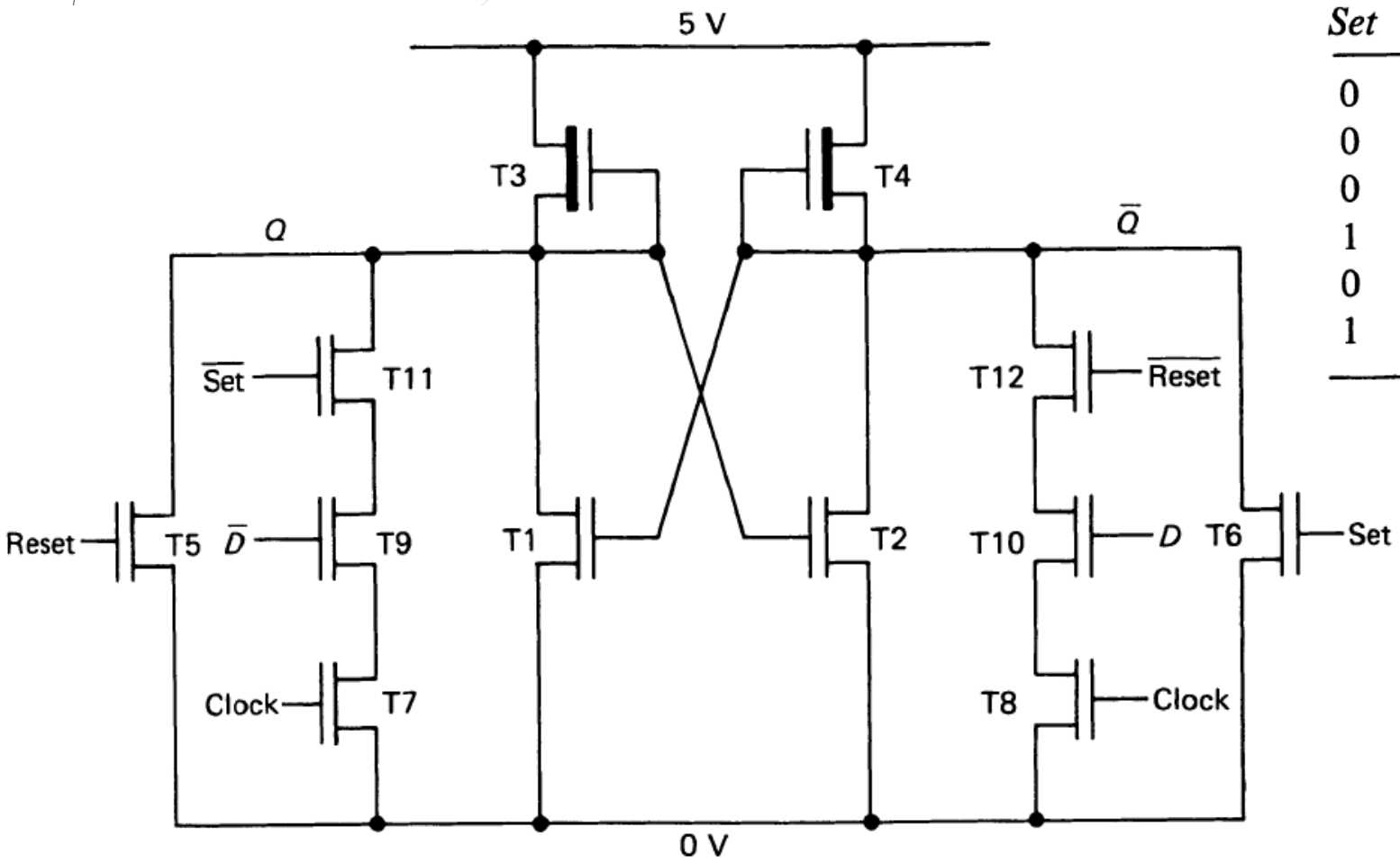


(c)

Figure 4.10 (NMOS) Set–Reset flip flop: (a) circuit diagram, (b) logic diagram, (c) logic symbol

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Indeterminate

D-type flip flop with overriding Set and Reset



Inputs				*Outputs*	
Set	*Reset*	*Clock*	*D*	*Q*	$\overline{Q}$
0	0	0	X	*Q*	$\overline{Q}$
0	0	1	0	0	1
0	0	1	1	1	0
1	0	X	X	1	0
0	1	X	X	0	1
1	1	X	X	0	0 – indeterminate

Inputs			*Outputs*		
J	*K*	*Clock*	*Q*	$\bar{Q}$	
0	0	1 → 0	*Q*	$\bar{Q}$	
1	0	1 → 0	1	0	
0	1	1 → 0	0	1	
1	1	1 → 0	1 → 0	0 → 1	Change state
			or 0 → 1	1 → 0	

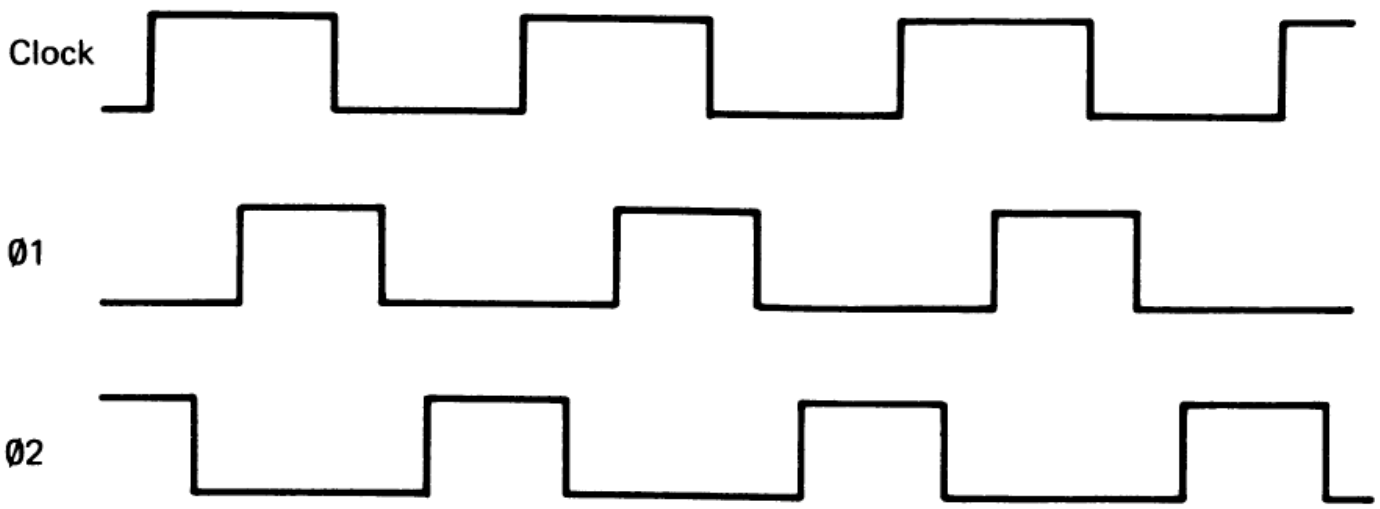


Figure 4.12 Two-phase non-overlapping clocks

Dynamic Flip Flops

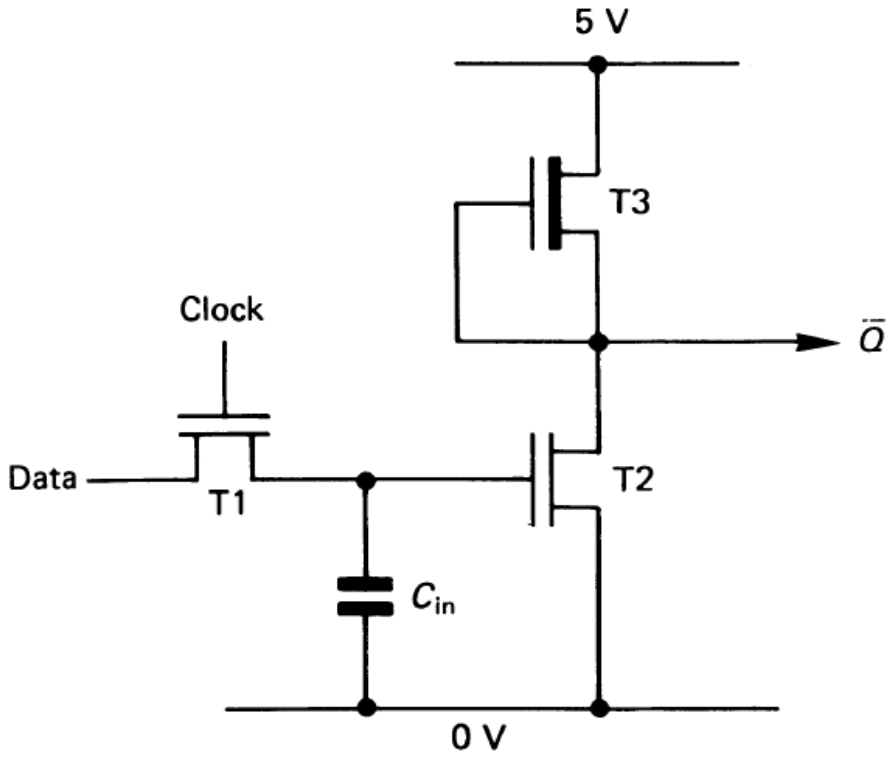


Figure 4.14 Basic dynamic flip flop

Table 4.4 Truth table for dynamic flip flop

Inputs		*Output*
Clock	*Data*	$\bar{Q}$
0	X	$\bar{Q}$
1	0	1
1	1	0

X = 0 or 1.

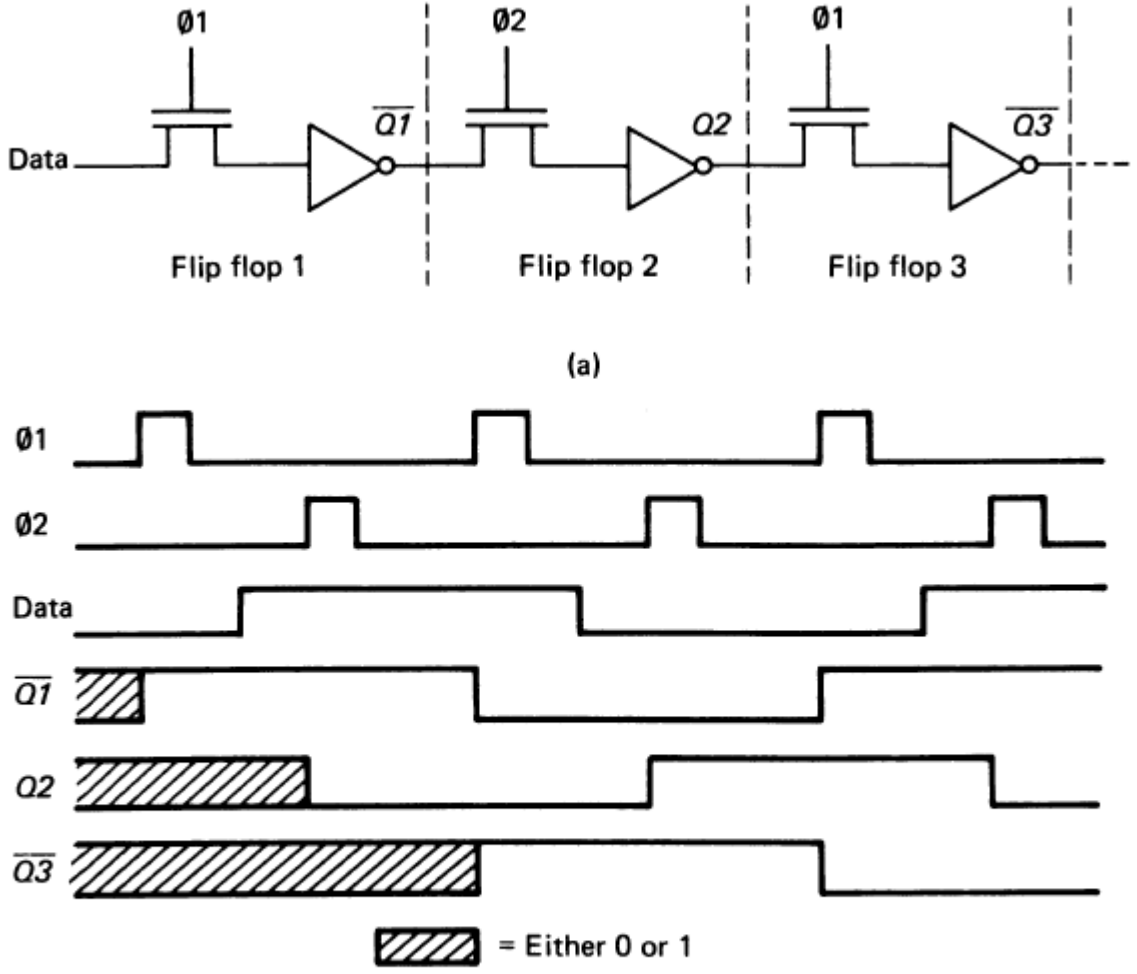


Figure 4.15 A shift chain using dynamic circuits: (a) circuit diagram, (b) timing

Data pipeline

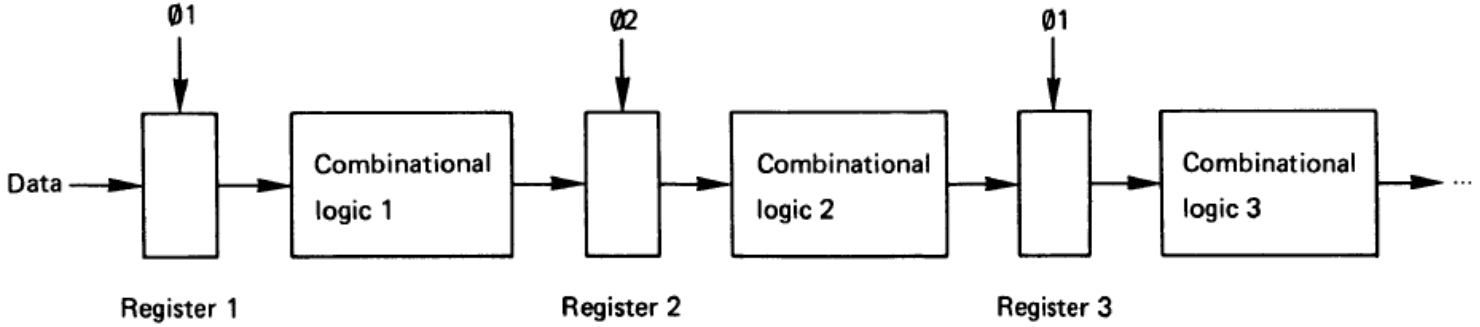


Figure 4.16 General schematic of a data pipeline

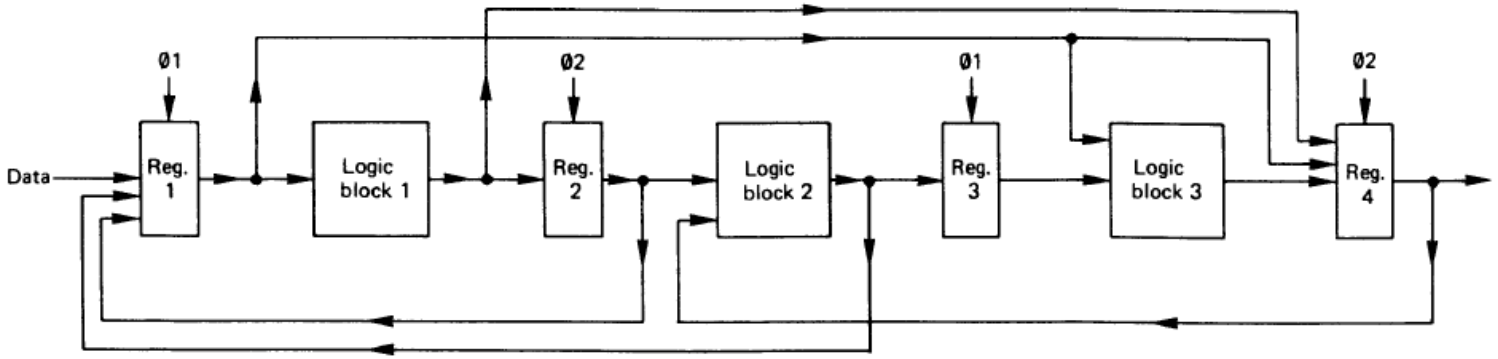
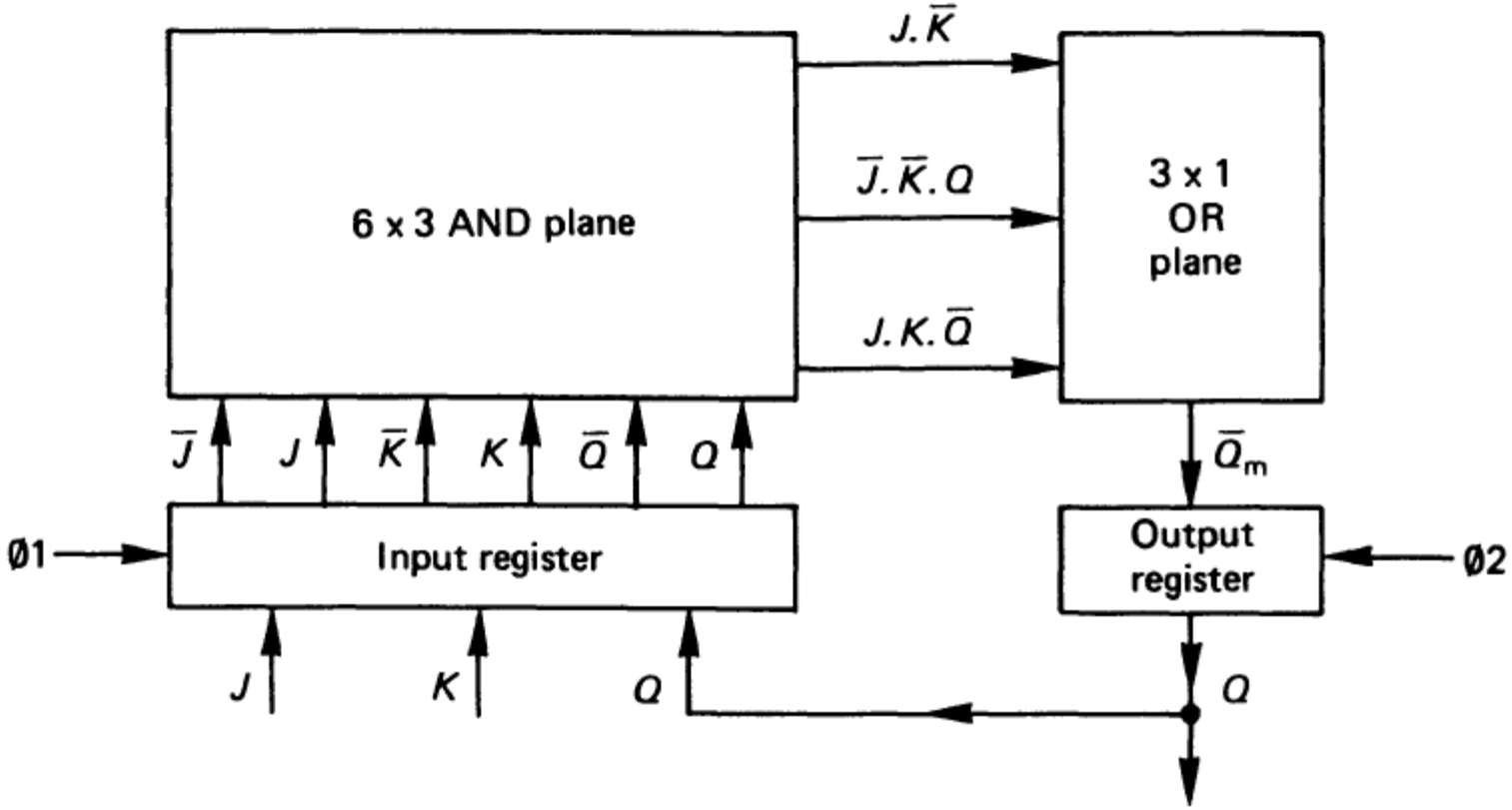


Figure 4.17 Examples of feedforward and feedback connections between pipeline stages

PLA based JK Flip Flop



(a)